Mapping the Narrow Corridor with Large Language Models

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July 1, 2026

Abstract

Acemoglu and Robinson's *The Narrow Corridor* traces national histories as paths through a two-dimensional (state power, society power) space but plots none: quantifying the two axes at each moment is the labor-intensive, judgment-laden task that has kept the framework qualitative. We ask whether large language models (LLMs) can operationalize it. We introduce a reproducible, provider-agnostic pipeline that scores a country period by period using chain-of-thought (events first, then score) and in-context anchoring on prior periods, against an explicit rubric with schema-validated output. As a proof of concept we produce a six-country trajectory atlas (Iran, France, the United Kingdom, the United States, China, Chile) and compare four models. We assess the scores for consistency with an expert index (V-Dem) and for inter-model agreement — reading agreement as concurrent consistency rather than accuracy, since such indices likely appear in the models' training data — and discuss conceptually how the biases of an LLM annotator and a human expert differ. Code, prompts, runs, and an interactive gallery of the animated trajectories are released.

1 Introduction

In *The Narrow Corridor* [2], Acemoglu and Robinson propose that liberty is neither the natural product of a strong state nor of a strong society, but of a contested balance between the two. They picture a country's history as a path through a plane whose horizontal axis is the power of civil society and whose vertical axis is the power of the state; a "narrow corridor" running between the axes marks the region where the two powers grow together and liberty is sustained. The book is rich in narrative but contains *no plotted trajectories.* The obstacle is not conceptual but operational: turning centuries of a nation's history into a sequence of $(\text{society}, \text{state})$ coordinates requires an enormous amount of expert judgment, and any such coding is vulnerable to the biases, selective readings, and coarse simplifications that historians rightly distrust.

This is a measurement problem, and measurement problems in the social sciences increasingly admit machine assistance. Large language models have been shown to annotate and classify political text at or above the level of crowd workers [11], and a growing literature examines their use as instruments in computational social science [19]. We ask a narrower, concrete question: *can an LLM place a country in the Narrow Corridor space, period by period, in a way that is reproducible, checkable against an established expert index, and useful to a researcher?*

We take an applied, digital-humanities stance. Our primary artifact is a **six-country trajectory atlas** — Iran, France, the United Kingdom, the United States, China, and Chile — rendered as time-shaded paths through the corridor, together with the cross-country patterns those paths reveal. The method and its validation are the backing that makes the atlas trustworthy rather than decorative. Concretely, we contribute:

1. **A scoring method.** A period-by-period pipeline combining an anchored 0–10 rubric, chain-of-thought elicitation of events and trends [18], and in-context anchoring on prior periods [5, 8].

2. **A provider-agnostic, open implementation.** Any model reachable through a unified interface can be swapped by changing one string; we release code, prompts, caches, and every run.

3. **A trajectory atlas and cross-country reading** of six countries.

4. **An interactive, publicly hosted web gallery.** A static site (GitHub Pages) presenting every trajectory as a time-shaded plot with *click-to-play animations* that reveal the path period by period and annotate the historical event driving each move; readers explore and compare models directly in the browser, turning a static figure into a navigable atlas.

5. **An assessment of the scores** for consistency with an expert index (V-Dem) and for inter-model

reliability, with explicit treatment of the training-data-overlap threat that makes such agreement concurrent consistency rather than proof of accuracy.

6. **A conceptual discussion of measurement bias**: what an LLM annotator plausibly gets right and wrong relative to a human researcher or an expert panel doing the identical task (we run no human study here; that comparison is left to future work).

2 Related Work

The Narrow Corridor and measuring state/society power. The (state, society) framing extends the institutional program of Acemoglu and Robinson [1, 2]. Quantifying the two axes connects to decades of effort to measure regime characteristics and state capacity: the Varieties of Democracy (V-Dem) project aggregates expert codings into latent indices including a core civil-society index and measures of state administrative capacity [6, 16]; Polity5 codes regime authority on an autocracy–democracy scale [14]; and Freedom House rates political rights and civil liberties [10]. These projects are themselves large expert coding operations, which makes them both our natural validation targets (Section 5.4) and our natural comparison point when we ask what an LLM changes about the coding process (Section 6).

Text as data and LLMs as social-science instruments. The use of automated methods to extract measures from text is well established [12]. Recent work reports that instruction-tuned LLMs match or exceed crowd annotators on political text-labeling tasks [11] and surveys their broader promise and pitfalls for computational social science [19]. A related strand uses LLMs to *simulate* human populations [3] and debates whether they can stand in for human participants at all [7]. Our task differs from labeling or simulation: we ask the model to produce a continuous, temporally coherent *measurement* of a theoretical construct, then check that measurement against independent data.

Prompting. Our pipeline relies on two now-standard techniques. Chain-of-thought prompting [18] improves reasoning by eliciting intermediate steps; we use it to force the model to enumerate historical events and trends *before* scoring. In-context learning [5, 8] lets the model condition on examples provided at inference time; we feed all prior periods' scores forward so the trajectory is temporally coherent rather than a sequence of independent guesses. This anchoring is a deliberate design choice with a cost — an early mis-score can propagate — which we return to in Section 7.

Bias and reliability. LLMs inherit biases from training data [4] and exhibit measurable political leanings: they do not reflect a representative cross-section of opinion [17], can carry the political slant of their pretraining corpora into downstream behavior [9], and have been probed directly for partisan bias [15]. These findings motivate both our reliability analysis — borrowing the logic of inter-coder reliability from content analysis [13] — and our discussion of how LLM bias compares to the well-known biases of human expert coders.

3 Method

The space. We place a country at each time period t at a point $(\mathrm{soc}_t, \mathrm{sta}_t) \in [0, 10]^2$, where soc_t is the power of civil society and sta_t is the power of the state. A period is a fixed span of years (e.g., a decade). We read a country as *in the corridor* in period t when its state and society power are close (near the diagonal) and *above/below* it when one clearly dominates. The dashed lines in our plots mark this band *illustratively*, near the diagonal, rather than as a fitted threshold, and are drawn at constant width — a deliberate simplification of the book's *widening* corridor; positions are therefore read qualitatively, not from a numeric cutoff.

Anchored rubric. A free-floating request for "two numbers" is not a measurement. Every scoring prompt embeds a fixed 0–10 rubric that defines each band of state power (from 0–2, collapsed/nonexistent, to 9–10, overwhelming penetration of society) and society power (from 0–2, atomized, to 9–10, society pervasively checks the state), plus explicit neutrality guidance: judge each period on its own terms, weigh evidence for and against a shift, avoid presentism, and treat the two axes as independent. The rubric is what makes scores comparable across periods, countries, and models.

Chain-of-thought: events then scores. For each period the model is first asked to list the major events and trends affecting state or society power, separating slow-moving trends from discrete events and noting the direction of each effect. This narrative is then inserted into a second prompt that requests the numeric score, so the number is conditioned on explicit reasoning rather than produced in one leap [18].

In-context learning: temporal anchoring. Starting from the earliest period (scored on absolute terms), we accumulate every prior period's $(\mathrm{soc}, \mathrm{sta})$ pair and feed the running trajectory into each subsequent scoring prompt [5]. The model reports both the change from the previous period and the resulting absolute values, which discourages the sequence from drifting and keeps a quiet period near zero change.

Structured, validated output. Each scoring call returns a JSON object — the two power values, their per-period changes, a short key event, and a one-sentence justification — validated against a fixed schema. A malformed response triggers one reminder retry and then a numeric fallback extractor before an error is raised.

Provider-agnostic access and caching. All model calls go through a single unified interface, so a model is selected by one identifier string and providers can be swapped without code changes. Because the pipeline makes one or two calls per period and is therefore expensive, every response is cached on disk keyed by (model, prompt); re-runs and visualization iteration are free, and the released cache makes our exact runs reproducible.

Visualization. A run renders as (i) a static plot of society vs. state power with the path time-shaded from early (dark) to late (bright) and a year colorbar, the largest moves annotated with their year and key event; and (ii) an animation that reveals the path period by period, drawing an arrow for the move each event caused and captioning it. Figure 1 is built from the static renderings; the animations are best viewed in the interactive gallery at `https://esonghori.github.io/visualize_narrow_corridor_with_ai/`.

4 Experimental Setup

Countries. We study six countries chosen to span regions of the corridor and of the world: **Iran (Persia)**, **France**, the **United Kingdom**, the **United States**, **China**, and **Chile**. Each is scored from a country-appropriate start year through 2020 (a common cutoff for consistency across cases) in decade (10-year) periods, giving 14–24 periods per country (Table 1 lists windows and counts). Six countries is a proof of concept, not a basis for cross-national generalization.

Models. We compare four models from four families (Google, Anthropic, OpenAI, Alibaba): three closed-weight models (`gemini-2.5-pro`, `claude-opus-4-8`, `gpt-5.5`) and one open-weight model (`qwen-2.5-72b-instruct`), which keeps an open- vs. closed-weight contrast. All are driven through the same interface with identical prompts, isolating model effects from prompt effects.

Consistency check against an expert index. We compare the LLM scores to V-Dem [6], aligning each period's midpoint year to the V-Dem year. We map LLM *society power* to the V-Dem core civil-society index (`v2xcs_ccsi`) and LLM *state power* to a state-authority measure (V-Dem state authority over territory, `v2svstterr`) — a narrow, territorial-control proxy for the rubric's broader penetration-of-society notion of state power. The mapping is fixed before inspecting correlations, and alternative capacity indicators are easy to swap in the released code. We report rank (Spearman) correlations per country on both *levels* and *first differences* (period-to-period change): levels are dominated by a shared secular trend and inflate correlation, so first differences are the more honest agreement signal. Crucially, V-Dem is itself almost certainly in the models' training data, so we read this as *concurrent consistency with expert consensus*, not as external proof of accuracy (Section 7). Polity5 and Freedom House are natural additional anchors but are out of scope here.

5 Results

5.1 Trajectory atlas

Figure 1 shows the six trajectories under `claude-opus-4-8`, chosen as a single reference model for legibility; the per-model panels for all four models are in the online gallery, and their disagreement is quantified in Section 5.5. The six paths span the corridor's regions and largely match the book's canonical placements. The **United Kingdom** stays inside the corridor throughout, state and society rising together (the Red Queen). **France** enters it, moving from society-leaning in 1789 to inside by the late twentieth century, and **Chile** loops out and back — a sharp fall of society and spike of state at the 1973 coup, then a return toward the corridor after the 1990 transition. **Iran** and **China** exit *upward* into state dominance (the Despotic Leviathan), Iran after 1979 and China after 1949. The **United States** is the outlier, drifting from a society-leaning founding toward state dominance — a placement we scrutinize below, as it departs from both the book and the expert data. Long diagonal runs mark co-growth; tight loops mark oscillation in contested periods; large single-period jumps mark ruptures.

5.2 Ensemble atlas (mean across models)

Figure 2 shows, for each country, the trajectory averaged across all four models period by period. Because independent models rarely share the same idiosyncratic error, the mean is a more robust reading than any single

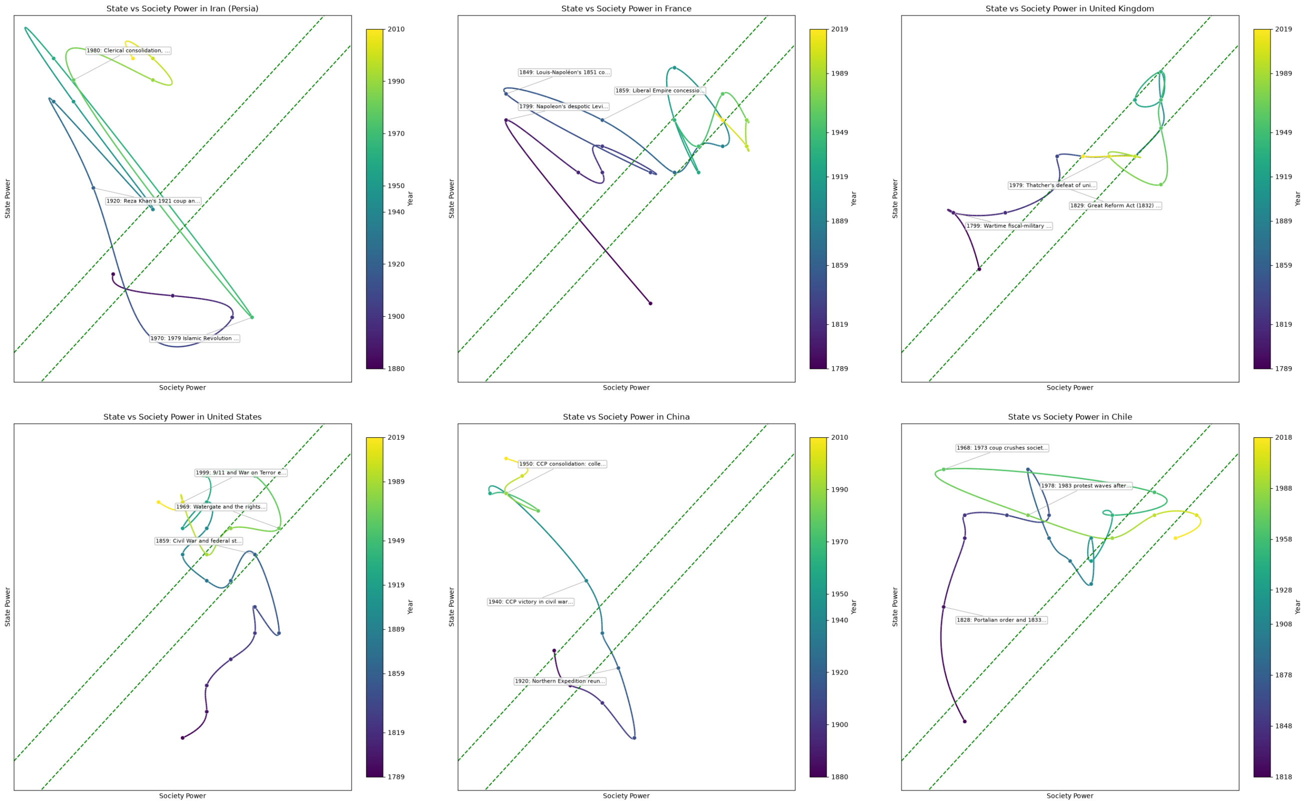

Figure 1: Trajectory atlas: society power (x) vs. state power (y) for six countries under `claude-opus-4-8`. Paths are time-shaded (dark = early, bright = late); dashed lines mark the corridor illustratively. Animated versions are available in the online gallery.

model's path: where the four agree it tracks the consensus tightly, and where they diverge it sits between them, so a point should then be read as uncertain rather than precise (the underlying per-axis agreement is quantified as Krippendorff's α in Section 5.5). We treat this mean atlas, not any single model, as the paper's headline reading. The mean is nearly identical to the reference model for the high-agreement cases (Iran, Chile; Section 5.5) and visibly smoother for the United States, where the models disagree most; the corridor placements are unchanged — UK inside, France and Chile in or entering it, Iran and China state-dominated, the US drifting toward state dominance.

5.3 Cross-country patterns

Three regularities recur. First, the countries the book treats as corridor-dwellers — the UK throughout, France and Chile by their late histories — trace long diagonal runs in which state and society grow together. Second, every country's largest single-period move is an institutional rupture that pushes the *state* axis up: the 1979 Iranian revolution, the 1973 Chilean coup, post-1949 Chinese consolidation, and the US Civil War. Third, the two twentieth-century revolutions from below (Iran, China) end in state dominance rather than liberty — precisely the Despotic Leviathan trap the framework predicts when a mobilized society confronts, but does not balance, a hardening state.

5.4 Consistency with V-Dem

Table 1 reports correlations between the LLM scores and V-Dem over overlapping years, on both levels and first differences. On *levels*, LLM society power tracks V-Dem's core civil-society index strongly for China ($\rho = 0.89$), France (0.87), Chile (0.74) and the UK (0.70); the striking exception is the United States ($\rho = -0.10$, discussed in the case studies). On the more demanding *first differences* the society signal is positive but weaker (China 0.66, Iran 0.60, France 0.40), as expected once the shared trend is removed. The *state* axis is where consistency collapses — first-difference $\rho \approx 0$ for four of six countries. Figure 3 shows why: our V-Dem state proxy (`v2svstterr`, authority over territory) is near-saturated and flat for consolidated states, so it cannot move with the LLM's broader state-*capacity* notion; the LLM tracks the *building* of the modern state that a territorial-control index does not. As noted above, this measures agreement with expert consensus the models

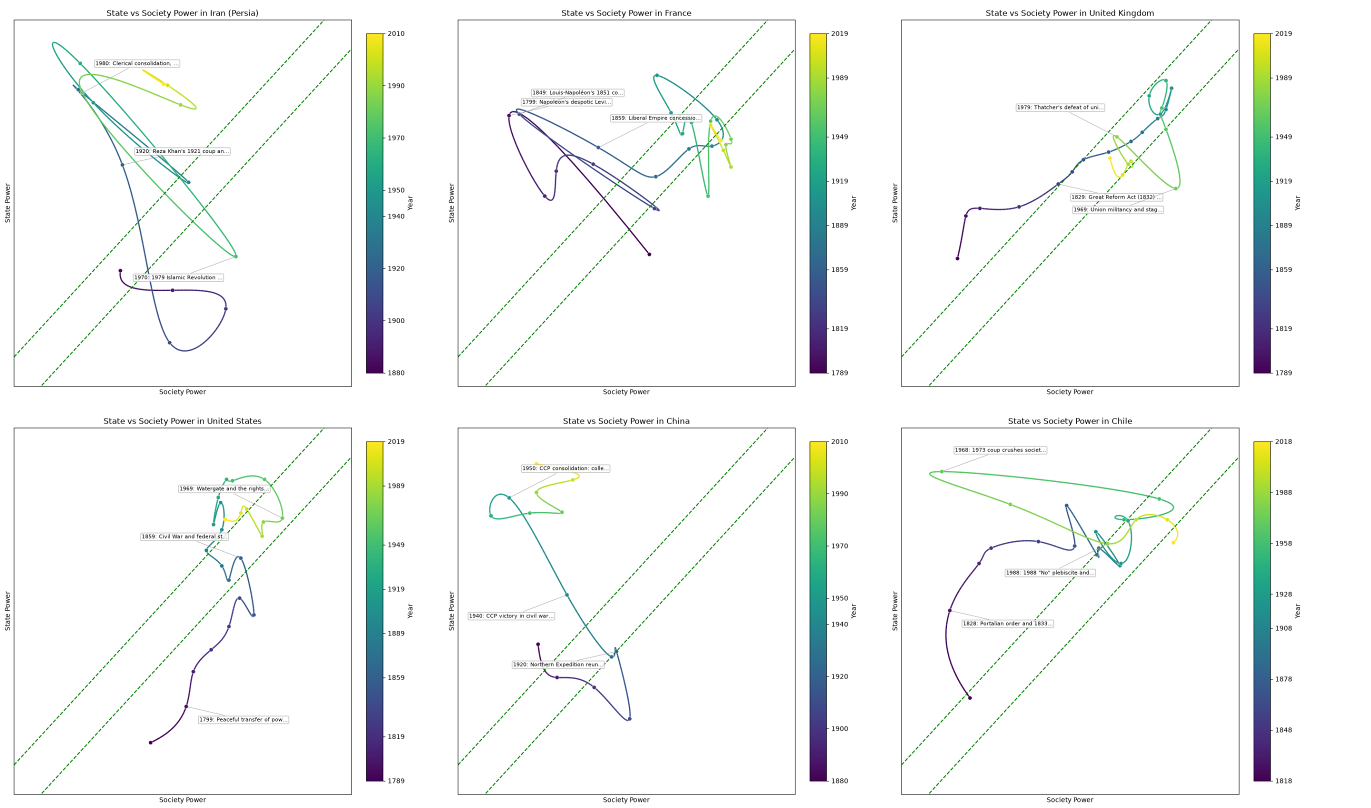

Figure 2: Ensemble atlas: the same six countries, each averaged across the four models period by period. Compared with the single-model atlas (Fig. 1), the mean damps any one model's idiosyncrasy; the spread it averages over is reported as Krippendorff's α in Table 2. Time-shaded (dark = early, bright = late); dashed lines mark the corridor illustratively.

were likely trained on, not independent accuracy.

Figure 3 re-plots each country from V-Dem in the same space (society = `v2xcs_ccsi`×10, state = `v2svstterr`/10), at the LLM period-midpoint years. The society axis broadly rhymes with the LLM ensemble (Fig. 2), but the state axis is pinned high for every consolidated state — V-Dem places even 1880s Qajar Iran and the early United States near the top of the state axis, the opposite of the LLM, which reads those as weak states still building capacity. The comparison is thus best read as *agreement on society, divergence on the operationalization of state power*.

5.5 Model comparison and reliability

Treating each model as a "rater" in the sense of intercoder reliability [13], Table 2 reports agreement between models on the same country. We report Krippendorff's α (interval, treating the 0–10 scores as interval-scaled and the four models as raters) on the period-to-period *changes* rather than on levels (for the same reason first differences are used in Section 5.4): agreement on changes tests whether models see the same *moves*, not merely the same secular trend. Models agree more on society ($\alpha = 0.74$ overall) than on the state axis (0.67), and most on the countries with the clearest arcs — Iran (0.77/0.78) and Chile (society 0.83). Agreement is lowest for the United States (society $\alpha = 0.47$), the same case whose atlas placement and V-Dem society correlation are most anomalous: model disagreement localizes exactly the contested reading, as a human panel's would. The open-weight model does not systematically diverge on the periods it scored; its real limitation was *reliability of structured output* — the strongest open model we tried, Qwen3-235B-A22B, could not reliably emit valid JSON on the longer prompts, so we fell back to the stable `qwen-2.5-72b-instruct` (Section 7).

5.6 Qualitative case studies

Iran, 1979. All four models place a society peak in the 1970s that collapses into state dominance in the 1980s (the ensemble society score falls sharply while state climbs past 8), matching the book's account of a mobilized society producing not liberty but a Despotic Leviathan. **Chile, 1973/1990.** The 1973 coup is the largest single move (society crashing, state spiking

Table 1: Consistency with V-Dem (`claude-opus-4-8`): Spearman ρ at period-midpoint years, reported as level / Δ (levels vs. first differences). Society vs. `v2xcs_ccsi`; state vs. `v2svstterr`. First differences are the more honest signal.

Country	Window	N	Society ρ	State ρ
Iran (Persia)	1880–2020	14	0.45 / 0.60	0.42 / 0.13
France	1789–2020	24	0.87 / 0.40	0.20 / 0.04
United Kingdom	1789–2020	24	0.70 / 0.21	0.13 / 0.07
United States	1789–2020	24	-0.10 / 0.26	0.84 / 0.15
China	1880–2020	14	0.89 / 0.66	0.74 / 0.24
Chile	1818–2020	21	0.74 / 0.17	0.18 / 0.42

Table 2: Inter-model reliability: Krippendorff's α (interval) on period-to-period *changes* across models, per country. Higher = models see the same moves.

Country	#models	Society α	State α
Iran (Persia)	4	0.77	0.78
France	4	0.72	0.56
United Kingdom	4	0.56	0.63
United States	4	0.47	0.62
China	4	0.56	0.56
Chile	4	0.83	0.57
Overall	–	**0.74**	**0.67**

near the ceiling), followed by recovery after the 1990 plebiscite — a country leaving and re-entering the corridor. **United States (a genuine divergence).** The models read a long slide from a society-leaning founding toward a strong federal and security state, ending state-leaning, with the Civil War as the largest move. This contradicts both the book, which treats the US as a corridor-dwelling Shackled Leviathan, and V-Dem, which keeps US civil society pinned high throughout (Fig. 3); it is also the country where the models agree least. We read it as a caution — the LLM appears to over-weight the growth of the federal and security state relative to the resilience of American civil society — rather than as a finding.

6 Discussion: LLM vs. Human Researcher or Expert Panel

Both an LLM and a human expert (or panel) performing this coding are biased; the useful question is *how the bias profiles differ*, and which task properties favor which coder.

What a human panel brings. Domain expertise, sensitivity to contested historiography, the ability to flag "this case is genuinely disputed," and accountable, attributable judgment. Established indices such as V-Dem manage individual coder bias with multiple experts per case and a latent measurement model that estimates and corrects for coder disagreement [16]; content analysis formalizes this as inter-coder reliability [13]. The costs are equally real: expert panels are slow, expensive, hard to scale to arbitrary country-periods, and carry their own systematic biases — national, ideological, disciplinary — that are difficult to audit because they live in individual judgment.

What an LLM changes. An LLM annotator is fast, cheap at scale, and *transparent in its instructions*: the rubric and prompts are explicit artifacts that can be inspected, criticized, versioned, and re-run identically. Its bias, however, is *opaque in origin*: it reflects the distribution of its training data [4, 9, 17], which over-represents English-language, recent, and Western sources, and it tends toward received consensus, likely flattening contested or under-documented episodes rather than flagging them. It has no accountable author, and its scores can shift with model version or provider (Section 5.5).

Reliability as the bridge. The inter-coder logic that human panels use for humans applies to models: agreement across independent models is a measurable proxy for reliability. Where independent models agree on the moves, a score is more likely to reflect a shared signal than one model's idiosyncrasy; where they disagree, the disagreement usefully localizes the contested cases a human panel would also flag. Reliability is necessary but not sufficient for validity, though — models trained on overlapping corpora can agree on a shared bias — which is why we treat both inter-model agreement and V-Dem consistency as evidence about *reliability and consensus*, not accuracy.

When to use which. We argue the LLM pipeline is best positioned as a *scalable first pass and a reproducible baseline* — generating candidate trajectories over many countries cheaply, surfacing where models disagree, and freeing expert attention for the contested cases — rather than as a replacement for expert coding where accountability and contested-case judgment are paramount.

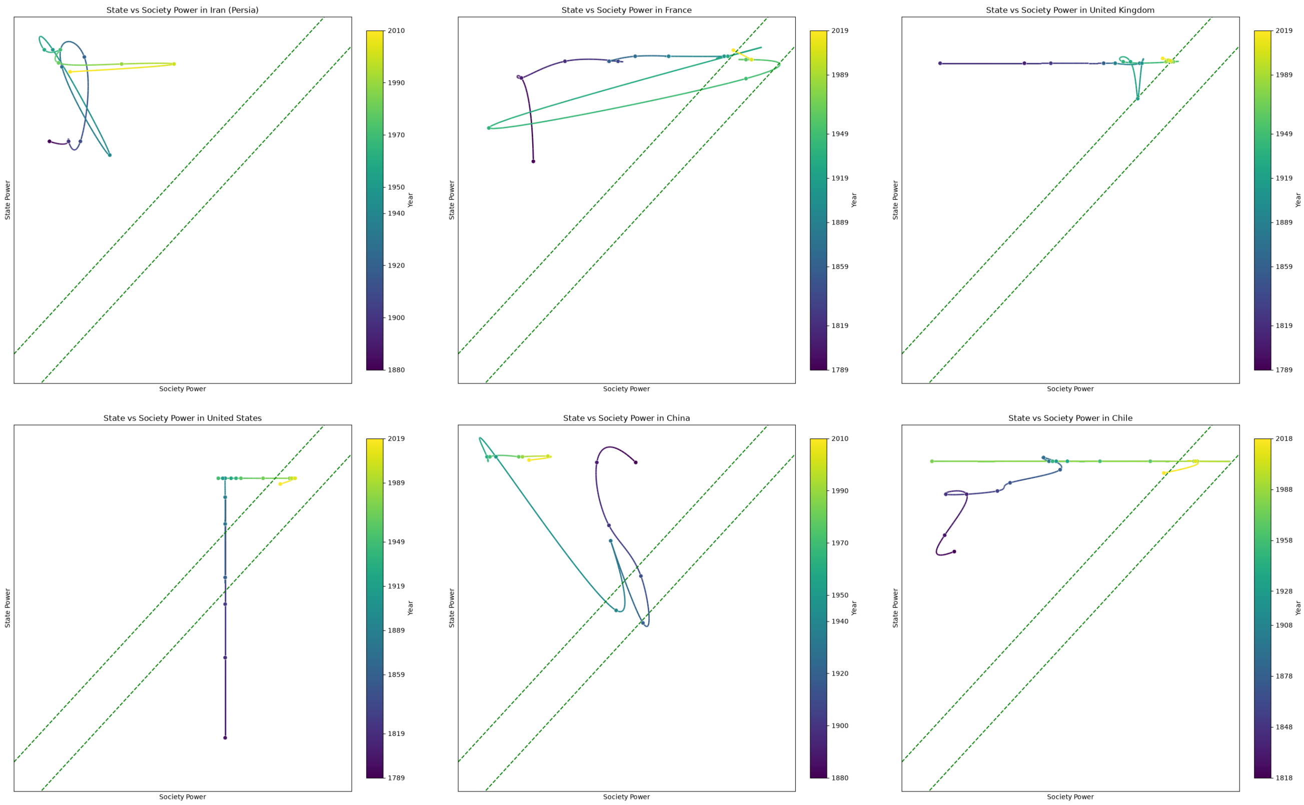

Figure 3: V-Dem atlas: the same six countries plotted from expert codings (society = core civil-society index $\times 10$; state = authority over territory $/10$) at the LLM period-midpoint years. Compared with the LLM ensemble (Fig. 2) the society axis broadly agrees, but V-Dem's territorial-authority state axis is near-saturated for consolidated states and misses the state-capacity dynamics the LLM captures.

7 Limitations

Several limitations bound what this proof of concept establishes. *Circularity of the consistency check.* V-Dem and similar indices are almost certainly in the models' training data, so agreement with them cannot distinguish measuring the construct from recalling memorized values; we therefore frame the check as concurrent consistency, not accuracy, and disentangling the two (e.g., counterfactual-history probes, or ablating the events step) is future work. *Construct compression.* The two axes reduce vast, contested constructs to single numbers, and our single state-capacity indicator only partially aligns with the theory's notion of state power. *Coverage and language bias.* Training corpora over-represent English-language, recent, and Western sources — weakest for exactly three of our cases (Iran, China, Chile), and we prompt in English throughout. *Path dependence.* In-context anchoring can propagate an early mis-score through the whole trajectory; we do not ablate it here. *Model idiosyncrasy.* We report the mean-across-models atlas (Fig. 2) as the headline reading and quantify per-axis spread (Section 5.5); still, all four are English-centric LLMs and may share bias. *Structured-output reliability.* The strongest open model we tried, Qwen3-235B-A22B, could not reliably emit valid JSON on the longer prompts (garbage or degenerate repetition), so we used the previous-generation `qwen-2.5-72b-instruct` — a practical caveat for open-weight reproduction. *Scale and statistics.* Six countries and short per-country series preclude cross-national generalization or strong significance claims. We deliberately keep the study small; the natural next steps — a human expert-coder baseline, prompt/period-granularity ablations, a non-English prompting probe, and more expert indices — are left to future work.

8 Conclusion

We presented, as a proof of concept, a method and a provider-agnostic tool that operationalize the Narrow Corridor's (state, society) space into reproducible trajectories, together with a protocol for checking their consistency with expert consensus and their reliability across models, and released it as a one-command exercise. The resulting six-country atlas is a first quantified rendering of a framework that has been purely qualitative, and the accompanying consistency check and bias discussion frame honestly what such LLM-generated measurements are — and are not — good for.

Reproducibility

All code, prompts, the on-disk response cache, and every run (JSON, static plots, animations) are released at https://github.com/esonghori/visualize_narrow_corridor_with_ai. The exact experiments in this paper are reproduced by `paper/experiments/run_experiments.py` and `analysis.py`; see `paper/README.md`. The interactive gallery of animated trajectories is hosted at https://esonghori.github.io/visualize_narrow_corridor_with_ai/.

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